One-Variable Statistics, Distributions & Margin of Error

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the botanical conservatory. What is the mean of the data set?

Value	Frequency
10	2
12	4
14	4
16	2

2 [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the regional bakery. What is the total number of measurements in the data set?

Value	Frequency
16	2
18	7
20	7
22	2

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3 [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the science museum.

What is the mean of the data set?

Value	Frequency
22	2
24	6
26	6
28	2

4 [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the ceramic studio.

What is the total number of measurements in the data set?

Value	Frequency
28	2
30	5
32	5
34	2

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5 [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the public transit depot. What is the mean of the data set?

Value	Frequency
34	2
36	4
38	4
40	2

6 [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the art gallery. What is the total number of measurements in the data set?

Value	Frequency
40	2
42	7
44	7
46	2

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7 [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the university lecture hall. What is the mean of the data set?

Value	Frequency
46	2
48	6
50	6
52	2

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the film archive, in a calculation for the orchestra hall (case 12), what is the mode of the data set: $44, 16, 10, 44, 45, 9, 12$?

9 [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the mountain observatory, in a calculation for the honeybee apiary (case 16), what is the mode of the data set: $42, 1, 2, 18, 11, 8, 2, 41$?

10 [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the recycling center, the numbers of books read by five students last month are: 6, 10, 14, 18, and 22. What is the mean number of books read?

11 [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the water treatment plant, the mean of four numbers is 18. Three of the numbers are 12, 16, and 22. What is the fourth number?

12 [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the regional airport shuttle, in a calculation for the coastal aquarium (case 28), if five quiz scores have a sum of 420, what is the mean score?

13 [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the glassblowing studio, in a class of 25 students, the median test score is 84. If 12 students scored above 84, how many students scored at or below 84?

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14 [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the ice skating rink, in a calculation for the wildlife sanctuary (case 36), what is the mode of the values: $12, 15, 18, 15, 21, 15, 18$?

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the ceramic studio, a data set of 9 numbers has a median of 15. If the number 15 is added to the data set, what is the new median?

16 [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

During a 5-day study at the botanical conservatory, the mean number of adult tickets recorded per day was 20. For the first four days, the counts were $15, 18, 21,$ and 22. What was the count on the fifth day?

17 [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

A random sample of visitors to the municipal library found that 36% prefer hardcover books, with an associated margin of error of $\pm 4\%$. If a total of 3000 visitors attend next month, what is the difference between the maximum and minimum plausible number of visitors expected to prefer hardcover books?

18 [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

At the community theater, Data Set A consists of the 5 measurements $34, 38, 42, 46, 50$, and Data Set B consists of the 5 measurements $24, 33, 42, 51, 60$. What is the positive difference between the range of Data Set B and the range of Data Set A?

19 [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

During a 5-day study at the regional bakery, the mean number of standard passes recorded per day was 23. For the first four days, the counts were $18, 21, 24,$ and 25. What was the count on the fifth day?

20 [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

A random sample of visitors to the coastal aquarium found that 39% prefer oak saplings, with an associated margin of error of $\pm 4\%$. If a total of 6000 visitors attend next month, what is the difference between the maximum and minimum plausible number of visitors expected to prefer oak saplings?

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the municipal library.

What is the total number of measurements in the data set?

Value	Frequency
12	3
14	5
16	5
18	3

a. 17

b. 15

c. 19

d. 16

22 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the community theater.

What is the mean of the data set?

Value	Frequency
14	4
16	6
18	6
20	4

a. 16

b. 20

c. 17

d. 18

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23 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the coastal aquarium. What is the mean of the data set?

Value	Frequency
18	3
20	4
22	4
24	3

a. 21

b. 22

c. 20

d. 24

24 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the mountain observatory. What is the total number of measurements in the data set?

Value	Frequency
20	4
22	5
24	5
26	4

a. 19

b. 17

c. 21

d. 18

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25 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the urban bike-share station. What is the total number of measurements in the data set?

Value	Frequency
24	3
26	7
28	7
30	3

a. 23

b. 20

c. 21

d. 19

26 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the solar microgrid facility. What is the mean of the data set?

Value	Frequency
26	4
28	4
30	4
32	4

a. 29

b. 30

c. 28

d. 32

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27 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the agricultural cooperative. What is the mean of the data set?

Value	Frequency
30	3
32	6
34	6
36	3

a. 32

b. 36

c. 33

d. 34

28 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the publishing house. What is the total number of measurements in the data set?

Value	Frequency
32	4
34	7
36	7
38	4

a. 25

b. 22

c. 23

d. 21

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

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29 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the ice skating rink, a two-way table of $N = 127$ observations on commuting mode records cell counts $u_1 = 30$ for (Junior, Train) and $v_2 = 35$ for (Senior, Bus). What is the diagonal sum $s = u_1 + v_2$?

a. 63

b. 65

c. 71

d. 68

30 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the meteorological station, a two-way table of $N = 88$ observations on commuting mode records cell counts $F_1 = 22$ for (Junior, Train) and $G_2 = 21$ for (Senior, Bus). What is the diagonal sum $w = F_1 + G_2$?

a. 41

b. 49

c. 46

d. 43

31 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the municipal library, in a calculation for the textile workshop (case 14), what is the range of the data set: $98, 30, 94, 76, 10, 11, 28$?

a. 88

b. 80

c. 95

d. 82

32 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the regional bakery, in a calculation for the geology lab (case 15), what is the mean of the data set: $7, 40, 31, 18$?

a. 23

b. 24

c. 21

d. 32

33 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the urban bike-share station, in a calculation for the carpentry workshop (case 17), what is the median of the data set: $45, 41, 59, 13, 65, 54, 68$?

a. 54

b. 57

c. 48

d. 45

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34 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the ceramic studio, in a calculation for the historic lighthouse (case 18), what is the mode of the data set: $33, 34, 27, 3, 4, 34$?

a. 40

b. 32

c. 34

d. 42

35 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the publishing house, data set A: $44, 47, 50$. Data set B: $35, 47, 59$. Which data set has the greater standard deviation?

a. Data set A

b. Data set B

c. They are equal

d. Cannot be determined

36 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the art gallery, what is the median of the following list of numbers: $3, 7, 12, 15, 19, 24, 30$?

a. 12

b. 15

c. 17

d. 19

37 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the robotics club, a biologist recorded the weights of six laboratory mice in grams: 14, 28, 19, 45, 12, and 33. What is the range of these weights?

a. 27

b. 31

c. 33

d. 45

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

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38 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the mountain observatory, a researcher wants to study the sleeping habits of all college students in a state. Which sample is most representative?

a. 50 students selected at random from the university library at midnight

b. 100 student athletes from one college

c. 500 students selected at random from across all colleges and universities in the state

d. 200 students who volunteer to fill out an online sleep survey

39 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the urban bike-share station, what happens to the standard deviation of a data set if 10 is added to every data point?

a. It increases by 10.

b. It increases by $\sqrt{10}$.

c. It remains unchanged.

d. It is multiplied by 10.

40 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the publishing house, a study randomly assigns 100 headache patients to receive either a new medication or a placebo. What type of study is this?

a. Observational study

b. Randomized controlled experiment

c. Census

d. Sample survey

41 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the recycling center, in a calculation for the local farmers market (case 70), why is random assignment used in experiments?

a. To ensure the sample represents the entire global population

b. To balance confounding variables across treatment groups so causality can be established

c. To eliminate the need for a control group

d. To reduce the sample size needed

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42 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

At the wildlife sanctuary, Data Set A consists of the 5 measurements $52, 56, 60, 64, 68$, and Data Set B consists of the 5 measurements $46, 53, 60, 67, 74$. What is the positive difference between the range of Data Set B and the range of Data Set A?

a. 12

b. 3

c. 28

d. 16

43 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

During a 5-day study at the water treatment plant, the mean number of fiction novels recorded per day was 41. For the first four days, the counts were $36, 39, 42,$ and 43. What was the count on the fifth day?

a. 41

b. 43

c. 48

d. 45

44 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

A random sample of visitors to the orchestra hall found that 37% prefer tulip bulbs, with an associated margin of error of $\pm 4\%$. If a total of 3000 visitors attend next month, what is the difference between the maximum and minimum plausible number of visitors expected to prefer tulip bulbs?

a. 1110

b. 290

c. 240

d. 120

45 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

At the sports complex, Data Set A consists of the 5 measurements $55, 59, 63, 67, 71$, and Data Set B consists of the 5 measurements $43, 53, 63, 73, 83$. What is the positive difference between the range of Data Set B and the range of Data Set A?

a. 16

b. 24

c. 6

d. 40

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